

Science Week 1: Gas exchange

Written application (20 to 30 minutes). Pen and paper, no notes. Year 8 Biology, B4 Organisms.

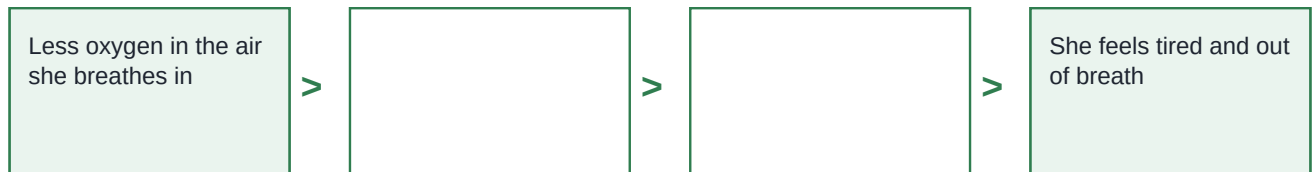
The one habit this week: the Because Chain

Every explain answer is a chain: **cause** → **effect** → **consequence**. Each link needs a joining word: *because, so, which means, therefore*. Finish by saying what the **body** gets or loses. A list of facts with no joining words stays at Level 1.

Task 1: Build a chain (8 minutes)

A climber goes high up a mountain, where the air has less oxygen in it. Explain why she soon feels tired and out of breath. (4 marks)

Fill the three empty boxes, then write the chain as full sentences below.



Task 2: Six-mark answer (15 minutes)

Explain how the structure of the lungs allows oxygen to reach the body's cells quickly. (6 marks)

Plan first: write three features in the margin, then give each a *so* or *which means* that leads to the next idea. End on respiration in the cells.

Mark yourself (tick one):

- ☐ **Level 1 (1 to 2):** features listed (thin walls, many alveoli) but not linked to an effect.
- ☐ **Level 2 (3 to 4):** some features linked to an effect, but the chain breaks or stops early.
- ☐ **Level 3 (5 to 6):** several features, each linked logically to the next, ending with the benefit to the cells.

Self-check (5 minutes)

- ☐ Each sentence leads to the next with a joining word (*so, because, which means*).
- ☐ No link is circular. I never wrote “this is good for gas exchange” as the reason.
- ☐ In Task 2, each feature has its own effect: thin walls → short distance; many alveoli → large surface area; capillaries → oxygen carried away.
- ☐ My last sentence says what happens to the cells or the body.
- ☐ I used some scientific words (diffuse, concentration, respiration), not just everyday ones.

Oral check with a parent (5 minutes)

Parent: read out only the *first* sentence of her Task 1 chain and ask: “**And then what happens?**” She continues the chain out loud. Repeat for each link. Then ask: “**Why does that cause the next thing?**” for the weakest link.

Answered without reading it ☐ Needed a prompt ☐ Could not answer ☐

Parent key

Task 1 model chain: Less oxygen in the air → the concentration of oxygen in the alveoli is lower, so there is a smaller difference between the alveoli and the blood → oxygen diffuses into the blood more slowly → less oxygen reaches the muscle cells, so they release less energy in respiration → she feels tired, and breathes faster to try to take in more.

Task 2 what to look for (Level 3): millions of alveoli → large surface area → more oxygen diffuses at once. Walls one cell thick → short diffusion distance → quick. Capillaries around each alveolus → oxygen carried away and the blood stays low in oxygen → diffusion carries on. Ventilation (breathing in and out) brings fresh air and removes carbon dioxide, which keeps the oxygen concentration high in the alveoli. All of it → oxygen reaches the cells quickly, for respiration. Level 2 answers list these without the ‘so’ links.

Weekly review (for J, 5 minutes)

Timings

Screen drill: _____ min Written: _____ min Self-check: _____ min

What was weak? (e.g. vague evidence, skipped working, skipped the check)

What worked?

Change for Week 2 (feeds the next build; next lesson in the B4 sequence is Breathing):